Newman-Penrose Formalism Calculations

Stephani et al., Equation (18.50)

Metric

$$ds^2 = \left(-\frac{\lambda r^p}{B^2 + r^{2\,p}}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(\frac{B\lambda p r^p x}{B^2 + r^{2\,p}}\right) \mathrm{d}t \otimes \mathrm{d}y + \left(\frac{B^2 + r^{2\,p}}{\lambda r^p}\right) \mathrm{d}r \otimes \mathrm{d}r + \left(\frac{1}{2}\left(B^2 r + r^{2\,p+1}\right) r^{-p-q}\right) \mathrm{d}x$$
$$\otimes \mathrm{d}x + \left(\frac{B\lambda p r^p x}{B^2 + r^{2\,p}}\right) \mathrm{d}y \otimes \mathrm{d}t + \left(-\frac{2\,B^2 \lambda p^2 r^{2\,p} x^2 - \left(B^4 r + 2\,B^2 r^{2\,p+1} + r^{4\,p+1}\right) r^q}{2\left(B^2 r^p + r^{3\,p}\right)}\right) \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (t, r, x, y)$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\sqrt{\lambda}\sqrt{\frac{r^p}{B^2 + r^{2\,p}}}\right) dt + \left(-\frac{1}{2}\sqrt{2} B \sqrt{\lambda} p x \sqrt{\frac{r^p}{B^2 + r^{2\,p}}} + \frac{1}{2}\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{\frac{B^2 r r^q}{r^p} + \frac{r r^{2\,p} r^q}{r^p}}\right) dy$$

$$n = \left(\sqrt{\frac{1}{2}}\sqrt{\lambda}\sqrt{\frac{r^p}{B^2 + r^{2\,p}}}\right) dt + \left(-\frac{1}{2}\sqrt{2} B \sqrt{\lambda} p x \sqrt{\frac{r^p}{B^2 + r^{2\,p}}} - \frac{1}{2}\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{\frac{B^2 r r^q}{r^p} + \frac{r r^{2\,p} r^q}{r^p}}\right) dy$$

$$m = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{\frac{B^2}{r^p} + \frac{r^{2\,p}}{r^p}}}{\sqrt{\lambda}}\right) dr + \left(\frac{i\sqrt{\frac{B^2}{r^p} + \frac{r^{2\,p}}{r^p}}\sqrt{r}}{2\,r^{\frac{1}{2}\,q}}\right) dx$$

$$\overline{m} = \left(\frac{\sqrt{\frac{1}{2}}\sqrt{\frac{B^2}{r^p} + \frac{r^{2\,p}}{r^p}}}{\sqrt{\lambda}}\right) dr + \left(-\frac{i\sqrt{\frac{B^2}{r^p} + \frac{r^{2\,p}}{r^p}}\sqrt{r}}{2\,r^{\frac{1}{2}\,q}}\right) dx$$

Contravariant components

$$l = \left(\frac{Bpr^p x}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}} - \frac{\sqrt{2}B^2}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} - \frac{\sqrt{2}r^{2\,p}}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} \right) dt + \left(\frac{r^{\frac{1}{2}\,p}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,q}} \right) dy$$

$$n = \left(-\frac{Bpr^p x}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}} - \frac{\sqrt{2}B^2}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} - \frac{\sqrt{2}r^{2\,p}}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} \right) dt + \left(-\frac{r^{\frac{1}{2}\,p}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,q}} \right) dy$$

$$m = \left(\frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{\sqrt{2}B^2 + \sqrt{2}r^{2\,p}} \right) dr + \left(\frac{i\,r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}} \right) dx$$

$$\overline{m} = \left(\frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{\sqrt{2}B^2 + \sqrt{2}r^{2\,p}} \right) dr + \left(-\frac{i\,r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}} \right) dx$$

Directional Derivatives

$$D = \frac{Bpr^p x}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}} - \frac{\sqrt{2}B^2}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} - \frac{\sqrt{2}r^{2\,p}}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}\,\partial_t + \frac{r^{\frac{1}{2}\,p}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,q}}\,\partial_y$$

$$\Delta = -\frac{Bpr^p x}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}} - \frac{\sqrt{2}B^2}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}} - \frac{\sqrt{2}r^{2\,p}}{2\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}\,\partial_t - \frac{r^{\frac{1}{2}\,p}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}r^{\frac{1}{2}\,q}}\,\partial_y$$

$$\delta = \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{\sqrt{2}B^2 + \sqrt{2}r^{2\,p}}\,\partial_r + \frac{i\,r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}}\,\partial_x$$

$$\overline{\delta} = \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{\sqrt{2}B^2 + \sqrt{2}r^{2\,p}}\,\partial_r - \frac{i\,r^{\frac{1}{2}\,p}r^{\frac{1}{2}\,q}}{\sqrt{B^2 + r^{2\,p}}\sqrt{r}}\,\partial_x$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = -\frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}pr^{\frac{1}{2}\,p}}{2\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}+\frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}-\frac{i\,\sqrt{B^2+r^{2\,p}}B\sqrt{\lambda}pr^{\frac{3}{2}\,p}}{\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}}+\frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}$$
$$+\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}pr^{\frac{5}{2}\,p}}{2\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}+\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}qr^{\frac{5}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}+\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}r^{\frac{5}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}pr^{\frac{1}{2}\,p}}{2\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}-\frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}+\frac{i\,\sqrt{B^2+r^{2\,p}}B\sqrt{\lambda}pr^{\frac{3}{2}\,p}}{\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}}-\frac{\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}$$
$$-\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}pr^{\frac{5}{2}\,p}}{2\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}-\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}qr^{\frac{5}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}-\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}r^{\frac{5}{2}\,p}}{4\left(\sqrt{2}B^4r+2\,\sqrt{2}B^2rr^{2\,p}+\sqrt{2}rr^{4\,p}\right)}$$

$$\alpha = \frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}pr^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}+\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}+\frac{i\,\sqrt{2}\sqrt{B^2+r^{2\,p}}B\sqrt{\lambda}pr^{\frac{3}{2}\,p}}{4\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}-\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}r^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}$$
$$-\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}pr^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}+\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}qr^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}-\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}r^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}$$

$$\beta = -\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}pr^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}-\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}-\frac{i\,\sqrt{2}\sqrt{B^2+r^{2\,p}}B\sqrt{\lambda}pr^{\frac{3}{2}\,p}}{4\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}+\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}B^2\sqrt{\lambda}r^{\frac{1}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}$$
$$+\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}pr^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}-\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}qr^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}+\frac{\sqrt{2}\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}r^{\frac{5}{2}\,p}}{8\left(B^4r+2\,B^2rr^{2\,p}+rr^{4\,p}\right)}$$

$$\tau = -\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r+\sqrt{2}rr^{2\,p}\right)}-\frac{\sqrt{B^2+r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r+\sqrt{2}rr^{2\,p}\right)}$$

$$\pi = \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}qr^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r + \sqrt{2}rr^{2\,p}\right)} + \frac{\sqrt{B^2 + r^{2\,p}}\sqrt{\lambda}r^{\frac{1}{2}\,p}}{4\left(\sqrt{2}B^2r + \sqrt{2}rr^{2\,p}\right)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{\lambda p^2 r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)} - \frac{\lambda q^2 r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)} + \frac{\lambda r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{\lambda p^2 r^p}{16\left(B^2r^2 + r^2r^{2\,p}\right)} - \frac{\lambda q^2 r^p}{16\left(B^2r^2 + r^2r^{2\,p}\right)} + \frac{\lambda r^p}{16\left(B^2r^2 + r^2r^{2\,p}\right)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{\lambda p^2 r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)} - \frac{\lambda q^2 r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)} + \frac{\lambda r^p}{8\left(B^2r^2 + r^2r^{2\,p}\right)}$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{\lambda p^2 r^p}{48\,(B^2 r^2 + r^2 r^{2\,p})} - \frac{\lambda q^2 r^p}{48\,(B^2 r^2 + r^2 r^{2\,p})} + \frac{\lambda r^p}{48\,(B^2 r^2 + r^2 r^{2\,p})}$$

Weyl Tensor Components

$$\begin{aligned}
\Psi_0 = {} & \frac{B^4 \lambda p^2 r^p}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda p q r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{B^4 \lambda q^2 r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{3i\,B^3 \lambda p^2 r^{2\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{i\,B^3 \lambda p q r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{3\,B^4 \lambda p r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{5\,B^2 \lambda p^2 r^{3\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{B^2 \lambda q^2 r^{3\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{3i\,B^3 \lambda p r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{3i\,B \lambda p^2 r^{4\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{i\,B \lambda p q r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{\lambda p^2 r^{5\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{\lambda p q r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{\lambda q^2 r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{3i\,B \lambda p r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^2 \lambda r^{3\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{3\,\lambda p r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{\lambda r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}
\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = {} & -\frac{B^4 \lambda p^2 r^p}{12\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda p q r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda q^2 r^p}{24\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{i\,B^3 \lambda p^2 r^{2\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{i\,B^3 \lambda p q r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda p r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{5\,B^2 \lambda p^2 r^{3\,p}}{6\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^2 \lambda q^2 r^{3\,p}}{12\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{i\,B^3 \lambda p r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{B^4 \lambda r^p}{24\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& + \frac{i\,B \lambda p^2 r^{4\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{i\,B \lambda p q r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{\lambda p^2 r^{5\,p}}{12\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{\lambda p q r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{\lambda q^2 r^{5\,p}}{24\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{i\,B \lambda p r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} \\
& - \frac{B^2 \lambda r^{3\,p}}{12\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{\lambda p r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{\lambda r^{5\,p}}{24\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}
\end{aligned}$$

$\Psi_3 = 0$

$$\Psi_4 = \frac{B^4 \lambda p^2 r^p}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda p q r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{B^4 \lambda q^2 r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$+ \frac{3i\,B^3 \lambda p^2 r^{2\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{i\,B^3 \lambda p q r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$- \frac{3\,B^4 \lambda p r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{5\,B^2 \lambda p^2 r^{3\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{B^2 \lambda q^2 r^{3\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$- \frac{3i\,B^3 \lambda p r^{2\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^4 \lambda r^p}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{3i\,B \lambda p^2 r^{4\,p}}{2\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$+ \frac{i\,B \lambda p q r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{\lambda p^2 r^{5\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{\lambda p q r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$- \frac{\lambda q^2 r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} - \frac{3i\,B \lambda p r^{4\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{B^2 \lambda r^{3\,p}}{4\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$
$$+ \frac{3\,\lambda p r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})} + \frac{\lambda r^{5\,p}}{8\,(B^6 r^2 + 3\,B^4 r^2 r^{2\,p} + 3\,B^2 r^2 r^{4\,p} + r^2 r^{6\,p})}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"